

Course Code: 203
Course Title: Operating System

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Course Title	Operating System									
Credits	4									
Course Category	Minor Course									
Level of Course	100-199 (Foundation / Introductory)									
Teaching per Week	4 Hours									
Minimum weeks per Semester	15 (Including class work, examination, preparation etc.)									
Review / Revision	2022-2023									
Implementation Year:	A.Y. 2023-2024									
Purpose of Course	An Operating System (OS) is a software that manages computer hardware and software resources and provides common services for computer programs. The operating system is an essential component of the system software in a computer system. Application programs usually require an operating system to function. The course is based on open source operating systems like Linux.									
Course Objective	4.To understand functionality provided by an Operating System. 5.To make aware with basic concepts of Windows O. S. Management. 6.To learn about device management.									
Pre-requisite	Basic knowledge of computers.									
Course Outcomes	CO1: Students will learn how operating system is important for computer system and what is the role of an OS, and also learn different types of operating system and their services. CO2: Students will be able to understand the structure and organization of file system. CO3: To differentiate between windows and linux OS CO4: To install and maintain linux workstation and also able to manage user accounts. CO5: To understand devices, usage of devices, scheduling algorithms and decide which is the best one. CO6: Students will be able to develop application the coordinate with respective OS in a much better way which is an essential.									
Mapping between Course Outcomes(CO) with Program Outcomes(PSO)			PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PSO7	PSO8
	CO1									
	CO2									
	CO3									
	CO4									
	CO5									
	CO6									
Course Content	Unit 1. Operating System Concepts 1.1.Evolution of Operating System & History 1.2.Need of an Operating System 1.3.Single User & Multi User Operating System 1.3.1 Types of OS and their advantages and dis-advantages 1.3.2 Batch OS, Distributed OS, Multi-Tasking OS 1.3.3 Rea-time OS, Mobile OS									

	1.4.Elements of an Operating System 1.5.Operating System as a Resource Manager Unit 2. Introduction to File System and File Management 2.1. File Concept 2.2. Operations on File 2.7. File Access Methods (Sequential Access and Direct Access) 2.8. Directory Systems File Management Functions. 2.9. File System and Directory Structure organization. 2.10.File Protection. Unit 3. Introduction of Linux 3.1.Introduction of Linux versions 3.2.Components of Linux 3.3.Comparison of Windows and Linux Unit 4. Linux Administration 4.3. Installing Linux 4.4. Installation of Open Source Software 4.3.Maintaining User Accounts 4.4.System Config Services (Package) Unit 5. Device Management 5.1.Device Management Function 5.2.Device Characteristics 5.3.Disk space Management 5.4.Allocation and Disk Scheduling Methods
Reference Books	11. Operating System Concepts: – James Peterson: – McGraw Hill 12. Operating System: – Stallings - PHI 13. Operating System Principles: – Silberschatz, Galvin, Gagne - Willey, India 14. Operating Systems – A. S. Godbole – Tata McGraw Hill 15. Linux – The Complete Reference – Richard Petersen – Tata McGraw Hill 16. "Operating System Concepts" Author: Abraham Silberschatz, Greg Gagne, Peter B. Galvin ISBN: 978-1118063330 Publisher: Wiley 17. "Linux System Programming: Talking Directly to the Kernel and C Library" Author: Robert Love ISBN: 978-1449339531 Publisher: O'Reilly Media 18. "Linux Bible" Author: Christopher Negus ISBN: 978-1118999875 Publisher: Wiley 19. "Understanding the Linux Kernel" Author: Daniel P. Bovet, Marco Cesati ISBN: 978-0596005658 Publisher: O'Reilly Media 20. "Linux Command Line and Shell Scripting Bible" Author: Richard Blum ISBN: 978-1118983843 Publisher: Wiley
Teaching Methodology	Class Work, Discussion, Self-Study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.